

1. FeNa <1% = prerenal

WHAT YOU SEE

Suspects in jail holding <1% signs

WHAT IT ENCODES

FeNa = $(\text{UNa} \times \text{PCr}) / (\text{PNa} \times \text{UCr}) \times 100$. <1% means avid tubular Na reabsorption from a healthy tubule responding to hypoperfusion = prerenal azotemia. >2% implies tubular injury (ATN). Validated only in oliguric AKI.

BOARD TRAP

Calling every low FeNa 'prerenal' — contrast nephropathy and early sepsis can give FeNa <1% from preserved tubular Na avidity despite intrinsic insult.

2. Early/intact tubular Na reabsorption

WHAT YOU SEE

'Early — intact tubular Na reabsorption' caption

WHAT IT ENCODES

Prerenal physiology: ↓ effective arterial volume → RAAS + ADH → maximal proximal and distal Na/water reabsorption. Hence low UNa (<20), high urine osm (>500), high BUN/Cr ratio (>20:1). The tubule itself is structurally intact.

BOARD TRAP

Confusing low UNa with low FeNa after diuretics — diuretics raise UNa/FeNa; use FeUrea instead.

3. Contrast can give <1%

WHAT YOU SEE

'CONTRAST <1%' suspect sign

WHAT IT ENCODES

Contrast-induced nephropathy classically produces a LOW FeNa (<1%) despite being intrinsic, because intense renal vasoconstriction mimics prerenal Na avidity. Cr peaks at 3–5 days. This is a key exception that breaks the FeNa rule.

BOARD TRAP

Picking ATN/high FeNa for contrast nephropathy — the discriminator is the low FeNa plus the 24–48h onset, 3–5 day peak timeline.

4. Rhabdo / myoglobin cast <1%

WHAT YOU SEE

'RHABDO — myoglobin cast' sign

WHAT IT ENCODES

Pigment nephropathy (rhabdomyolysis) is another intrinsic cause that can show FeNa <1% early because vasoconstriction and proximal Na avidity dominate before frank ATN. Look for CK >5000, dipstick blood without RBCs, ↑K, ↑PO4, ↓Ca.

BOARD TRAP

Trusting low FeNa to exclude tubular injury here — muddy brown / pigmented casts plus dipstick-blood-negative-RBC is the real discriminator.

5. Sepsis early <1%

WHAT YOU SEE

'SEPSIS <1%' sign in jail

WHAT IT ENCODES

Early sepsis can give FeNa <1% (prerenal phase from vasodilation/hypovolemia) before evolving into septic ATN with FeNa >2% and muddy brown casts. Timing matters: the same patient migrates across categories.

BOARD TRAP

Anchoring on a single FeNa value — serial FeNa plus sediment is needed; a rising FeNa with muddy casts signals transition to ATN.

6. FeUrea when on diuretics

WHAT YOU SEE

'USE FeUrea instead' running diuretic

WHAT IT ENCODES

FeUrea = $(\text{Uurea} \times \text{PCr}) / (\text{BUN} \times \text{UCr}) \times 100$. Use when patient is on loop/thiazide diuretics that artificially raise FeNa. FeUrea <35% = prerenal; urea reabsorption is less diuretic-dependent than Na.

BOARD TRAP

Reporting FeNa >1% as 'ATN' in a diuretic-treated patient — FeUrea <35% reveals the true prerenal state.

7. CKD baseline raises FeNa

WHAT YOU SEE

Officer with 'CKD BASELINE' sign

WHAT IT ENCODES

Chronic kidney disease patients have a chronically elevated baseline FeNa (each nephron excretes more Na) so FeNa loses prerenal specificity. Interpret cautiously and lean on clinical volume status and sediment.

BOARD TRAP

Misreading an elevated FeNa in CKD as acute tubular necrosis — the elevation is chronic and expected.

8. Pigment nephropathy: heme+, no RBCs

WHAT YOU SEE

Corkboard 'PIGMENT NEPHROPATHY — heme but no RBCs'

WHAT IT ENCODES

Myoglobin and hemoglobin make the urine dipstick positive for blood while microscopy shows NO red cells. This dissociation is pathognomonic for pigment nephropathy (rhabdo or hemolysis) causing tubular injury.

BOARD TRAP

Diagnosing glomerulonephritis on a positive dipstick — absence of RBCs on micro rules out hematuria and points to myoglobin/hemoglobin.

9. Hematuria needs RBCs (not pigment)

WHAT YOU SEE

'HEMATURIA — NOT THIS' red X

WHAT IT ENCODES

True hematuria requires intact RBCs on microscopy. Dysmorphic RBCs/RBC casts = glomerular; isomorphic = post-renal/urologic. Pigmenturia (heme+, RBC-negative) is explicitly NOT hematuria.

BOARD TRAP

Working up 'hematuria' urologically when micro shows no RBCs — that's pigmenturia, work up rhabdo/hemolysis instead.

10. S3 proximal tubule = most vulnerable

WHAT YOU SEE

'S3 proximal tubule — most vulnerable' crime scene

WHAT IT ENCODES

The S3 segment (outer medulla) and medullary thick ascending limb sit in a low-O2 watershed, making them the first to die in ischemic/toxic ATN. 'Broken pipe' = sloughed brush border → muddy brown granular casts.

BOARD TRAP

Locating ischemic ATN damage in the cortex/glomerulus — the outer medullary S3 segment is the lesion site.

11. Muddy brown casts → ATN

WHAT YOU SEE

'MUDDY BROWN — ATN' footprint card

WHAT IT ENCODES

Pigmented (muddy brown) granular casts and renal tubular epithelial cells are the sediment signature of ATN. Their presence shifts the diagnosis from prerenal to intrinsic regardless of a low FeNa.

BOARD TRAP

Calling muddy brown casts 'normal granular casts' — they specifically indicate tubular epithelial necrosis (ATN).

12. WBC casts → AIN

WHAT YOU SEE

'WBC CAST — AIN' lantern card

WHAT IT ENCODES

WBC casts (especially with eosinophiluria, fever, rash, recent drug like PPI/NSAID/beta-lactam) indicate acute interstitial nephritis. Sterile pyuria with WBC casts and no bacteria supports AIN over UTI.

BOARD TRAP

Attributing WBC casts to pyelonephritis only — drug-induced AIN gives WBC casts with eosinophils and a culprit medication.

13. RBC casts → GN / vasculitis

WHAT YOU SEE

'RBC CAST — GN/vasculitis' hammer card

WHAT IT ENCODES

RBC casts and dysmorphic RBCs are diagnostic of glomerulonephritis / vasculitis. Pair with proteinuria, hypertension, ↓C3/C4 or positive ANCA to localize the glomerular lesion.

BOARD TRAP

Dismissing RBC casts as contamination — they are a glomerular red flag mandating serologies and often biopsy.

14. Hyaline casts → prerenal/benign

WHAT YOU SEE

'HYALINE — prerenal' ghost card

WHAT IT ENCODES

Hyaline casts are acellular Tamm-Horsfall protein, seen in concentrated urine, volume depletion, exercise, or diuretics. They support a prerenal/benign picture and are not specific for injury.

BOARD TRAP

Reading bland hyaline casts as evidence of intrinsic disease — they actually favor prerenal azotemia with intact tubules.